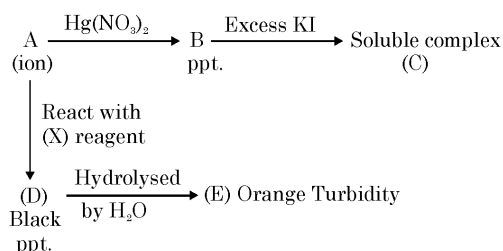


SCQ (Single Correct Type) :
1.


What is the hybridisation of central atom of (C) & what is the compound (E).

- (A) dsp^2 and HgI_2 (B) sp^3d^2 and $BiOCl$ (C) d^2sp^3 and BiI_3 (D) sp^3 and $BiOI$

2.

 Which of the following statement is/are **incorrect** ?

- (I) Both graphite and inorganic graphite are slippery in nature.
 (II) Both graphite and inorganic graphite are conducting electricity but graphite is less conducting in nature compared to inorganic graphite
 (III) Graphite is not having any charge separation like inorganic graphite
 (IV) All atoms in graphite as well as inorganic graphite are sp^2 hybridized.
 (V) $Li > Na > K > Rb > Cs$ (Reducing power in gaseous state)
 (VI) $Li > Na > K > Rb > Cs$ (Magnitude of enthalpy of atomization)
 (VII) $Li < Na < K < Rb < Cs$ (softness)
 (VIII) $Li > Na > K > Rb > Cs$ (strength of bond in their metal lattice)
- (A) I, III, IV (B) II, V, VII (C) V, VII, VIII (D) VI, VII, VIII

3.

The value of K_p for the reaction at $27^\circ C$ $Br_2(l) + Cl_2(g) \rightleftharpoons 2BrCl(g)$ is '1atm'. At equilibrium in a closed container partial pressure of $BrCl$ gas is 0.1 atm and at this temperature the vapour pressure of $Br_2(l)$ is also 0.1 atm. Then what will be minimum moles of $Br_2(l)$ to be added to 1 mole of Cl_2 , initially, to get above equilibrium situation:

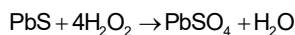
- (A) $\frac{10}{6}$ moles (B) $\frac{5}{6}$ moles (C) $\frac{15}{6}$ moles (D) 2 moles

4.

In 0.1 M $NaOH$ solution, equal volume of 0.1 M HCl is added, the conductivity of solution becomes $0.005 \Omega^{-1} cm^{-1}$. A further addition of HCl solution, the volume of which is equal to that of the first portion added, the conductivity increases to $0.017 \Omega^{-1} cm^{-1}$ then $\wedge_m(HCl)$ is :

- (A) $410.15 \Omega^{-1} cm^2 mol^{-1}$ (B) $4.1515 \Omega^{-1} cm^2 mol^{-1}$
 (C) $830.3 \Omega^{-1} cm^2 mol^{-1}$ (D) $8.303 \Omega^{-1} cm^2 mol^{-1}$

5. (i) Boiling point of H_2O_2 is greater than water due to stronger hydrogen bonding in H_2O_2 as compared to water.
 (ii) It restores the color of the old paints blackened by the action of H_2S . The lead sulphide (black) gets oxidized to lead sulphate (white)



Black (White)

(iii) H_2O is formed by the oxidation of 2-Ethylanthraquinone

(iv) H_2O_2 is a polar and non-polar

Which of the following set is of correct statements:

- (A) (i) and (ii) (B) (ii) and (iv) (C) (i) and (iii) (D) (i), (ii) & (iv)

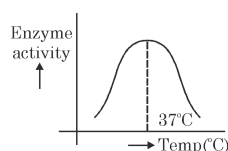
6. Consider a reaction $3\text{A}(\text{g}) \rightarrow 4\text{B}(\text{g}) + \text{C}(\text{g})$, starting with pure A having pressure $\frac{3}{5}$ atm, the pressure after 10 min. reaches to 1 atm. Calculate the value of rate of disappearance of A at initial instant.

- (A) $0.06 \text{ atm min}^{-1}$ (B) $0.6 \text{ mole lit}^{-1} \text{ min}^{-1}$ (C) $\frac{3}{50} \ln \frac{3}{5} \text{ atm min}^{-1}$ (D) $0.36 \text{ atm min}^{-1}$

MCQ (Multiple Correct Type) :

7. Which is / are correct options ?

(A) For Enzyme catalyst



- (B) Surfactants decreases the surface tension of liquid on dissolving in it.
 (C) Conductance of oil in water emulsion increase on addition of common salt into it.
 (D) At CMC, molecules associate to form particles of colloidal dimension.

8. For a reaction : $\text{CH}_3\text{COCH}_3 + \text{Br}_2 \xrightarrow{[\text{H}^+]} \text{CH}_3\text{COCH}_2\text{Br} + \text{H}^+ + \text{Br}^-$ following data was obtained. From the data, identify the correct statement(s).

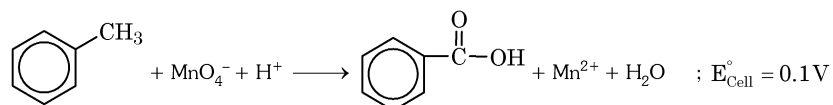
[Acetone]	[Br ₂]	[H ⁺]	[Rate of Reaction]
0.15	0.25 M	0.025 M	$6 \times 10^{-4} \text{ M sec}^{-1}$
0.15	0.50 M	0.025 M	$6 \times 10^{-4} \text{ M sec}^{-1}$
0.15	0.25M	0.050M	$12 \times 10^{-4} \text{ Msec}^{-1}$
0.2	0.25 M	0.025 M	$8 \times 10^{-4} \text{ M sec}^{-1}$

- (A) The overall order of the reaction is 2nd order
 (B) During the reaction, where $[\text{H}^+]$ does not change it will follow first order kinetics.
 (C) When $[\text{Acetone}] \gg [\text{Br}_2]$ then reaction will proceed with variable rate of most of the reaction
 (D) Order of reaction w.r.t Br_2 will be 1.
9. If Aufbau's principal and Hund's rule were not followed:
- (A) 'K' would have been d-block element & paramagnetic.
 (B) 'Cu' would have been s-block element.
 (C) 'Cr' would have been diamagnetic.
 (D) 'K' would have been diamagnetic.

10. Select **INCORRECT** statement(s):
- (A) pK_a value of B(OH)₃(aq.) increases on adding glycerol.
- (B) Aqueous solution of Ca(OH)₂ can not be used for removal of temporary hardness of water.
- (C) Crystals of borax on heating does not produce any acidic oxide.
- (D) Two layers of solid orthoboric acid are mainly connected together by hydrogen bonds.

11. 1 M solution of a dibasic acid H₂A (K_{a1} = 10⁻⁶ and K_{a2} = 10⁻¹⁰) is titrated against 1 M NaOH solution. Choose the correct statement A regarding the relative concentration of H₂A, HA⁻ and A²⁻ at various pH during titration.
- (A) At pH = 3 ; [H₂A] is highest among all
- (B) At pH = 6.5 ; [HA⁻] is highest among all
- (C) At pH = 10 ; [H₂A] is least among all
- (D) At pH = 8 ; [A²⁻] is highest among all

12. The sign of ΔG in a process can predict the feasibility of the process provided that the process occurs at constant pressure & temperature & there is no non P - V work involved. For a reaction:



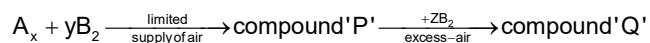
Then identify the incorrect statements.

- (A) At standard conditions, 65.9 kJ of electrical work can be obtained per mole of toluene consumed.
- (B) At standard conditions, 40.25 kJ of electrical work can be obtained per mole of KMnO₄ consumed.
- (C) At standard conditions, 32.17 kJ of electrical work can be obtained per mole of H₂SO₄ consumed.
- (D) At standard conditions, 40.25 kJ of electrical work can be obtained per mole of MnSO₄ produced.
13. A 10 mL mixture containing 0.2 g NaOH and 0.25g Na₂CO₃ was treated with xNHCl required 15 mL of it with phenolphthalein indicator. After end point a 0.5 g NaHCO₃ was added and it was again titrated with y N HCl required 25.5 mL of HCl with methyl orange. Which of the following is/are correct?
- (A) x is almost 1.5 times of y
- (B) If in above titration x N HCl is replaced with y N HCl then for first end point it will require approx. 23mL of HCl.
- (C) After first end point instead of 0.5 g if 2.5 g NaHCO₃ is added and titrated with x N HCl it will require approx. 65.5 mL.
- (D) Only (A) and (B) are correct.
14. The correct statement is/are
- (A) B₂H₆ acts as a lewis acid and also contain 3c-2e⁻ bonds.
- (B) BeCl_{2(s)} exist in polymeric form with sp² hybridised Be atom.
- (C) BI₃ acts as strong lewis acid due to poor back bonding.
- (D) The ionisation energy of O⁻ is higher than that of S⁻.
15. Find correct statements for Coordination compounds.
- (A) [PtCl₂(NH₃)₂] and [Cu(NH₃)₄]²⁺ are square planar complexes.
- (B) In [Fe(C₂O₄)₃]³⁻, geometrical isomerism does not exist while optical isomerism exists
- (C) CO bond order : [Ni(CO)₄] > [Co(CO)₄]⁻ > [Fe(CO)₄]²⁻.
- (D) [Ni(Cl₂)(PPh₃)₂] has two geometrical isomers.

16. Identify true statements.

- (A) Aquatic species are more comfortable in cold water because O_2 has more solubility at low temperature.
- (B) Henry's constant ' K_H ' (atm) increases with increase in temperature for gas like O_2 , N_2 .
- (C) Solubility of gas increases with increase in partial pressure of gas in water.
- (D) Solubility of gas increases with increase in temperature in water.

17. Consider the following reactions:



If atomic number of elements A and B are 15 and 8 respectively, then according to the given information the correct statement(s) is/are:

- (A) (B-A-B) bond angle in compound 'Q' > (B-A-B) bond angle in compound 'P'.
- (B) (A-B) bond length in compound 'Q' < (A-B) bond length in compound 'P'.
- (C) Value of $x + y + z$ is 9.
- (D) Compound 'P' is P_2O_5 and compound 'Q' is P_4O_{10} .

Match the column

18. Match the Matrix.

Column – I (Pairs of reaction)	Column-II (Related observation in each pair of reaction)
(P) $Cr^{3+}(aq.) \xrightarrow{\text{Excess } Na_2O_2} \text{Product (X)}$ $Cr^{3+}(aq.) \xrightarrow{\text{Excess } Na_2O} \text{Product (Y)}$	(1) X is soluble but Y is precipitate
(Q) $Al^{3+}(aq.) \xrightarrow{\text{Excess NaOH}} \text{Product (X)}$ $Pb^{2+}(aq.) \xrightarrow{\text{Excess NaOH}} \text{Product (Y)}$	(2) X and Y both are soluble
(R) $S^{2-} \xrightarrow{BaCl_2} \text{Product (X)}$ $SO_3^{2-} \xrightarrow[H_2O_2]{BaCl_2} \text{Product (Y)}$	(3) X and Y show different observation in the test tube
(S) $CrO_4^{2-} \xrightarrow{BaCl_2 / CH_3COOH} \text{Product (X)}$ $CrO_4^{2-} \xrightarrow{CaCl_2 / CH_3COOH} \text{Product (Y)}$	(4) Y is soluble but X is precipitate

Codes:

	P	Q	R	S
(A)	(1)	(2)	(3,4)	(3)
(B)	(2,3)	(2)	(1,3)	(3,4)
(C)	(2)	(3,4)	(3,4)	(1)
(D)	(1)	(2,3)	(1)	(4)

(Numerical Value type) :

19. A metal 'M' having atomic mass 31.25 crystallizes in cubic close packing and it shows "Schottky defects". If the edge length of the cubic lattice is 500 pm and density of the metal is 1.6075 gm/ml then calculate number of moles of metal atom 'M' missing per litre of the crystal.

Given: $1 \text{ amu} = 1.67 \times 10^{-24} \text{ gm}$

20. An element X (atomic wt. = 125) crystallizes in simple cubic structure. If diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. Find the density of pure element X in gm/cm^3 .

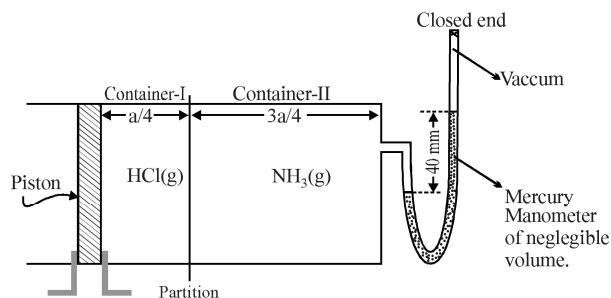
[Given : $\sqrt{3} = 1.732$, $N_A = 6 \times 10^{23}$]

[Fill your answer by multiplying it with 300.]

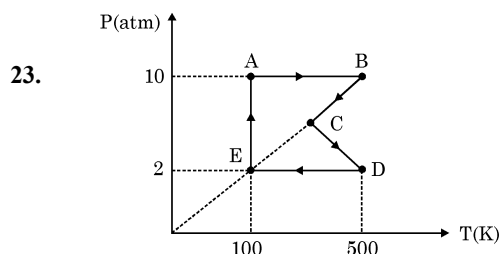
21. (A) $\text{Co}^{2+} + \text{KNO}_2 + \text{CH}_3\text{COOH} \longrightarrow$ Yellow ppt. complex with CN = X
 (B) $\text{Ni}^{2+} + \text{DMG} \longrightarrow$ Red ppt. with Y – number of O–Ni–O bond angle(s).
 (C) $\text{Cu}^{2+} + \text{K}_4[\text{Fe}(\text{CN})_6] \longrightarrow$ Total Z – number of metal ions in one ppt compound unit.
 (D) $\text{Al}^{3+} + (\text{NH}_4)_2\text{S} \xrightarrow{\text{H}_2\text{O}}$ Number of Sulphur atoms in ppt is P.

Report value of (X + Y + Z + P) ?

22. In the system shown below container-I contains $\text{HCl}(\text{g})$ at pressure 20 mm of Hg and container-II contains only $\text{NH}_3(\text{g})$ both at temperature 367.2 K.



If the partition is removed and piston is moved by "a/2" distance to the right side, then the height (in mm) difference in manometer after the complete reaction of HCl and NH_3 [$\text{HCl}(\text{g}) + \text{NH}_3(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$] is



For 1 mole of ideal gas, consider the cyclic process ABCDEA as shown in the graph through 5 steps (A – B, B – C,....., E – A).

Select number of correct statements

(i) At least 3 states have same volume.

(ii) For every step $\frac{V_{\text{final}}}{V_{\text{initial}}} \neq 1$

(iii) $V_{\text{maximum}} = V_B$

(iv) Heat inflow occurs in 3-steps.

- (v) Heat outflow occurs in 2-steps.
- (vi) Among all steps, maximum $|\Delta U|$ value is observed in $A \rightarrow B$ and $D \rightarrow E$ processes.
- (vii) $|Q_{BC}| = |Q_{CD}|$
- (viii) $\Delta U_{BC} = \Delta_{CD}$

24. How many correct statements in the following?

- (i) Phosphoric acid is more acidic than phosphinic acid.
- (ii) Hypo phosphorus acid can reduce Ag^+ into Ag .
- (iii) Boiling point order of element of 16th group is $Te > Po > Se > S > O$
- (iv) SF_4 is gas while SeF_4 & TeF_4 are solid.